Research Objectives

Skin Cancer Detection Using Deep Learning

Systematic 11-Step Analysis

Based on 39 Peer-Reviewed Research Papers

Research Topic: Skin Cancer Detection Using Deep Learning
Dataset: HAM10000 (10,015 dermoscopic images, 7 classes)
Methods: CNN, Transfer Learning, Data Augmentation, Grad-CAM
Course: Research Methodology — BCA 7th Semester
Protocol: Evidence-Bound, Zero Fabrication, IEEE Standards

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Generated: September 15, 2026

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1. Step 1 — Detailed Analysis of All 39 Papers

Each of the 39 research papers was analyzed across 16 dimensions. Where information was not available, "*Not Reported*" is indicated. Due to volume, a representative subset of papers is presented in full detail below; the complete data for all 39 papers is available in the accompanying `STEP_01_Paper_Analysis.md` file.

1.1 Paper 1: Skin Lesion Classification Using CNN With Novel Regularizer

Authors: M. A. Albahar **Year:** 2019

Dimension	Extracted Data
Research Problem	Discriminating between benign and malignant skin lesions from dermoscopic imagery
Research Aim	Develop CNN with novel weight-controlling regularizer ($\lambda = 0.02$)
Main Objectives	*Not Reported*
Dataset	ISIC archive (23,906 images; subsets of ∼8,000)
Lesion Classes	Nevus, melanoma, seborrheic keratosis, BCC, solar lentigo
Architecture	Custom CNN: 2 conv layers, max pooling, dropout, 128-neuron dense
Preprocessing	Resizing to $300 \times 300 \times 3$; gamma correction
Augmentation	*Not Reported*
Training	70/30 split; 100 epochs
Metrics	Accuracy 97.49%; AUC 0.77–0.93 (binary)
Findings	High binary accuracy with novel regularizer
Limitations	Binary pairwise only; no multi-class; no multi-center validation
Future Work	Unified multi-class classification
Research Gaps	Need for multi-class unified classification
Recommendations	*Not Reported*

1.2 Paper 3: Interpretable Skin Cancer Classification (HAM10000)

Authors: K. Mridha et al. **Year:** 2023

Dimension	Extracted Data
Research Problem	Need for interpretable models in healthcare applications
Research Aim	Develop optimized CNN with Grad-CAM/Grad-CAM++ interpretability
Dataset	HAM10000 (10,015 images; 7 classes)
Architecture	Custom CNN with batch normalization, max pooling, dropout
Preprocessing	Image resizing; pixel normalization
Augmentation	Random rotations, zooms, shear transforms, flips
Metrics	Accuracy 82.0%; Specificity 94.17%; Sensitivity 82.0%
Limitations	Modest minority-class sensitivity; needs clinical validation
Research Gaps	Class imbalance handling; interpretability improvement

1.3 Paper 7: Systematic Benchmark of Deep Learning for Melanoma

Authors: A. Naeem et al. **Year:** 2020

Dimension	Extracted Data
Research Problem	Lack of standardized evaluation protocols across dermatological datasets
Research Aim	Systematic benchmark of popular CNN architectures
Dataset	PH2, ISIC 2016, ISIC 2017, ISIC 2018, HAM10000
Architecture	ResNet-50, VGG-16, Inception-v3, DenseNet-121
Preprocessing	Aspect-ratio preserving resizing; hair suppression; color normalization
Augmentation	Horizontal/vertical flipping; zooming
Metrics	Accuracy up to 93.4%; Sensitivity 91.0%; Specificity 94.2%; AUC 0.95
Limitations	Class imbalance inflates accuracy; no clinical metadata integration
Research Gaps	Severe class imbalance; standardized evaluation protocols needed

1.4 Paper 12: Performance Enhancement Using Computer Vision

Authors: A. Magdy et al. **Year:** 2023

Dimension	Extracted Data
Research Problem	Enhancing feature quality before deep learning classification
Research Aim	Combine CV preprocessing (DullRazor + CLAHE) with transfer learning
Dataset	HAM10000 (10,015 images; 7 classes); ISIC 2019
Architecture	DenseNet-201, ResNet-50, MobileNetV2 with fine-tuning
Preprocessing	Dull Razor hair removal; CLAHE contrast enhancement
Augmentation	Rotations; flipping
Metrics	Accuracy 97.75%; Precision 97.50%; F1 97.35%; AUC 0.985
Limitations	Processing latency; not validated on smartphone images
Research Gaps	Synergy between preprocessing and deep learning

1.5 Paper 21: IncepX-Ensemble on HAM10000

Authors: S. Chatterjee et al. **Year:** 2024

Dimension	Extracted Data
Research Problem	Single-model representation bottlenecks in multi-class classification
Dataset	HAM10000 (10,015 images; 7 classes)
Architecture	InceptionV3 + Xception ensemble with concatenation fusion
Augmentation	Rotation, zooming, shifts, flipping
Metrics	Accuracy 96.87%; Precision 96.90%; F1 96.87%; AUC 0.982
Limitations	High parameter count; inference latency; no external validation
Research Gaps	Ensemble efficiency vs. accuracy trade-off

Note on Complete Analysis

The complete 16-field structured analysis for **all 39 papers** is contained in the companion file `STEP_01_Paper_Analysis.md` (50.7 KB, 782 lines). Due to length constraints, the remaining 34 papers follow the identical structure shown above.

2. Step 2 — Objective-Relevant Literature Matrix

Table 6: Literature Matrix: All 39 Research Papers

#	Title	Yr	Method/Model	Dataset	Key Metric	Limitations	Research Gap	Relevance	Rel.
1	Skin Lesion Classification CNN Novel Regularizer	19	Custom CNN	ISIC (23.9K)	Acc 97.49%	Binary only	Multi-class need	Transfer learning ref	H
2	ROI Transfer Learning Framework	20	AlexNet, VGG-16	DermIS, DermQuest	Acc 97.9%	Binary; hair issues	Multi-class; artifacts	TL comparison	H
3	Interpretable CNN Optimized (HAM10K)	23	Custom CNN + Grad-CAM	HAM10000	Acc 82.0%	Low minority sens.	Imbalance; XAI	Grad-CAM ref	H
4	Ensemble Lightweight DL	20	U-Net + Ensemble	ISBI 2016/17	Acc 87.3%	Binary; margin issues	Lightweight multi-class	Ensemble ref	M
5	DeepSkin Classification	23	Custom CNN + ResNet	HAM10000	Acc 88.0%	mel/bkl confusion	Class overlap	TL comparison	H
6	FCN-DenseNet Framework	20	FCN + DenseNet	ISIC 2017/18	Acc 95–98.5%	High computation	Dual-stage overhead	Preprocessing	M
7	Melanoma DL Benchmark	20	ResNet, VGG, etc.	Multiple ISIC	Acc 93.4%	Imbalance inflates	Standard protocols	Benchmark ref	H
8	Combined Decision Learners	22	VGG+ResNet+CapsNet	ISIC (2.8K)	Acc 93.50%	Binary; overhead	Ensemble multi-class	Ensemble ref	M
9	FixCaps Improved	22	FixCaps (CapsNet)	HAM10000	Acc 96.49%	Minority drops	Minority class CapsNet	HAM10K bench	M
10	Deep Clustering COMF-TL	22	VGG/ResNet + COMF	ISIC 2018/19	BAcc 83.7%	High train time	Feature imbalance	Imbalance ref	H
11	Neural Architecture Search	20	DARTS NAS	HAM10000	BAcc 78.4%	GPU overhead	Auto-architecture	HAM10K bench	M
12	CV Enhancement Preprocess	23	DenseNet-201 + CLAHE	HAM10000	Acc 97.75%	Latency	Preprocess synergy	Preprocessing	H
13	EFAM-Net Feature Fusion	24	EFAM-Net attention	HAM10000	Acc 93.85%	GPU memory	Attention for minority	Attention ref	H
14	Hair Segmentation U-Net	21	U-Net FCN	ISIC/PH2 (196)	Px Acc 95.8%	Latency; smoothing	Artifact removal	Preprocessing	L
15	StyleGAN Decision Fusion	20	StyleGAN + Ensemble	HAM10000	Acc 92.83%	GAN overhead	GAN vs. trad. aug	Augmentation	M

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#	Title	Yr	Method/Model	Dataset	Key Metric	Limitations	Research Gap	Relevance	Rel.
16	Joint Learning System	23	Multi-stream CNN	ISIC Archive	Acc 98.60%	Binary; params	Joint multi-class	Multi-stream ref	M
17	Multiple Connected NNs	20	U-Net + Parallel CNNs	MED-NODE (170)	Acc 97.50%	Tiny dataset	Multi-class larger data	Small ref	L
18	DermCDSM Clinical Support	24	RF, SVM + ResNet	DermNet (5K+)	Acc 91.00%	Variable lighting	ML vs DL comparison	ML/DL ref	M
19	Swin Transformer Hybrid	24	Swin + DGS-NLAN	HAM10000	Acc 94.21%	GPU demands	Transformer–CNN	Transformer ref	H
20	IoMT Intelligent Things	19	Lightweight NN	Clinical (1.2K)	Acc 91.20%	Heuristics fail	DL for IoMT	Edge ref	L
21	IncepX-Ensemble TL	24	InceptionV3 + Xception	HAM10000	Acc 96.87%	High params	Ensemble efficiency	TL benchmark	H
22	IAN-GDN Ensemble	20	ResNet+DenseNet+Inc	ISIC 2017	Acc 91.50%	3 classes only	Ensemble extension	Ensemble ref	M
23	AlexNet+SVM Pipeline	24	AlexNet + SVM	ISIC/HAM (3K)	Acc 98.66%	Outdated AlexNet	Modern architectures	Baseline ref	L
24	NCC k-Means Segmentation	24	NCC + k-means	PH2, ISBI 2016	Seg Acc 99.29%	No classification	Seg+classification	Segmentation ref	L
25	MIL 3D Features	22	CNN + 3D + MIL	50 biopsy cases	Acc 84.00%	3D equipment	Impractical for std.	Low	L
26	SPCB-Net Multi-Scale	24	SPCB-Net SIAP	HAM10000/ISIC	Acc 93.60%	GPU memory	Multi-scale modeling	Architecture ref	H
27	IARS SegNet	24	SegNet + Attention	ISIC 2017/18	Seg Acc 96.80%	Seg only	Seg + classification	XAI ref	M
28	Ear Disease Multimodal	23	ResNet + 1D CNN	Endoscopy (2.4K)	Acc 94.90%	Not dermatology	Multimodal concept	Methodology ref	M
29	Quantum Chebyshev	24	DWT + Autoencoder	HAM10000	Acc 98.23%	No backprop	Novel features	Feature ref	L
30	Knowledge Graph Melanoma	26	CNN + KG	Derm7pt, HAM	AUC 88.60%	Manual ontology	Neuro-symbolic	XAI ref	H
31	YOLOv9 + Faster R-CNN	25	YOLOv9 + F-RCNN	ISIC 2019/20	Acc 95.80%	Memory; latency	Object detection	Detection ref	M
32	VLM Report Generation	25	ViT + Lang. Decoder	ISIC (4K+)	F1 90.25%	Hallucination risk	VLM reports	Future ref	H

Continued on next page

#	Title	Yr	Method/Model	Dataset	Key Metric	Limitations	Research Gap	Relevance	Rel.
33	Multi-Scale DCNN PCE Loss	24	Multi-branch DCNN	HAM10000	Acc 91.80%	Branch latency	Dynamic loss imbalance	Loss function ref	H
34	NMF Segmentation	25	NMF + MI	Dermofit (1.3K)	Seg Acc 93.80%	Low overlap	Unsupervised seg	Low	L
35	Siamese CBHIR	25	Siamese Network	Histopathology	Top-1 88.50%	Biopsy reliance	Retrieval interpret.	XAI concept	M
36	FoMoSkinNet Dual-Stream	26	FocalMod + CNN	ISIC 2024 (12K+)	Acc 99.46%	Param footprint	Non-dermoscopic	Architecture ref	H
37	MSPE-DeiT Multimodal	26	DeiT + MLP meta	HAM10000	Acc 91.76%	Transformer cost	Metadata fusion	Multimodal ref	H
38	MoLE VQA	26	LLaVA + LoRA	ISIC/HAM	VQA 79.40%	Prototype only	VQA dermoscopy	Future ref	H
39	SLGAN-EID Enhancement	26	Dual-disc. GAN	HAM10000	Acc 94.30%	Mode collapse	Feature-preserving	Enhancement ref	M

Relevance Key: H = High, M = Medium, L = Low. Relevance assessed against project scope (HAM10000, CNN/Transfer Learning, BCA-level).

3. Step 3 — Research Problem Identification

3.1 What Problem Are Researchers Solving?

Researchers are primarily developing automated deep learning systems for the detection and classification of skin cancer from dermoscopic images. The overarching clinical goal is objective, early-stage lesion screening to assist—not replace—dermatologists.

3.2 Commonly Used Approaches

- **Dataset:** HAM10000 is the most widely used benchmark (16 of 39 papers)
- **Architectures:** Transfer learning with ResNet-50 (9 papers), VGG-16/19 (6 papers), DenseNet (5 papers), InceptionV3 (4 papers)
- **Preprocessing:** Hair removal via DullRazor [6,12,16,23,27,39]; CLAHE [12]; resizing to 224×224 [5,8,13,16,19,26,28,36]
- **Augmentation:** Geometric transforms in 15+ papers; GAN-based synthesis [15,39]

3.3 Established Research Areas

1. Use of large-scale public datasets (HAM10000, ISIC archives)
2. Application of transfer learning using standard architectures
3. Standard geometric data augmentation and basic preprocessing
4. Binary and multi-class formulations for dermoscopic classification

3.4 Common Limitations (Evidence-Based)

Recurring Limitations Across 39 Papers

1. **Class imbalance inadequately addressed** — Papers [3,5,7,9,10,11,13,33]. Melanocytic nevi comprise ∼67% of HAM10000.
2. **Binary classification only** — Papers [1,2,4,8,16,17] despite multi-class clinical need.
3. **No multi-center/clinical validation** — Papers [1,2,3,5,7,9,12,15,18,19,21,37,39].
4. **High computational overhead** — Papers [6,8,10,11,13,15,19,25,26,29].
5. **No model interpretability** — Majority of papers lack explainability.
6. **Confusion between similar classes** — mel vs. bkl [3,5].
7. **Metrics limited to accuracy only** — Noted by Paper [7].

3.5 Recurring Research Gaps

Identified Research Gaps

1. Severe class imbalance in HAM10000 not adequately resolved [3,7,9,10,11,13,33]
2. Limited systematic comparison of transfer learning architectures on HAM10000 [5,7,12]
3. Insufficient preprocessing pipeline evaluation [6,12,14,39]
4. Most studies lack explainability—only Paper [3] uses Grad-CAM on HAM10000

5. Few studies systematically evaluate data augmentation strategies [7,15,39]
6. Comprehensive multi-metric evaluation rarely performed [7]

3.6 Gaps Realistically Addressable Using HAM10000

1. Implementing targeted augmentation or class weighting for class imbalance
2. Systematic comparison of standard transfer learning architectures
3. Evaluating preprocessing pipeline impact on classification
4. Applying Grad-CAM to higher-performing transfer learning models
5. Comprehensive multi-metric evaluation across all 7 classes

4. Step 4 — Objective Theme Identification

Table 7: Literature-Supported Objective Themes

Theme	Papers	Gap/Opportunity	Literature Finding	Project Rel.
Image Preprocessing	[1,6,7,9,10,12,13,16,19,23,26,27,28,36,39]	Pipeline evaluation [6,12,14]	DullRazor + CLAHE yields 97.75% [12]	**High**
Data Augmentation	[2,3,5,6,7,8,9,12,13,14,15,16,19,21,22,26,36,39]	Synthetic augmentation evaluation [7,15,39]	Standard geometric widely used; GANs emerging	**High**
Transfer Learning	[1,2,3,5,6,7,8,10,12,15,16,18,20,21,22,28,39]	Limited comparison on HAM10000 [5,7,12]	ResNet, DenseNet, VGG dominant	**High**
Multi-class Classification	[3,5,9,11,12,13,15,19,20,29,30,33,37,38,39]	Binary vs. multi-class [1,2,4,8,16,17]	7-class HAM10000 classification	**High**
Class Imbalance	[3,5,7,9,10,11,13,15,21,33,39]	Rarely addressed; frequently cited limitation	Nevi ∼67%; minority <3%	**High**
Explainable AI	[3,27,30,35]	Only Paper [3] uses Grad-CAM on HAM10000	Significant gap	**High**
Model Evaluation	[7]	Metrics limited to accuracy	Multi-metric needed	**High**

5. Step 5 — Candidate Research Objectives

Based on the identified research gaps and literature analysis, 8 candidate objectives are proposed:

Objective 1: Preprocessing Evaluation

To evaluate the impact of standardized image preprocessing pipelines (including hair removal via DullRazor and contrast enhancement via CLAHE) on the multi-class classification performance of deep learning models on the HAM10000 dataset.

Gap Addressed: Insufficient preprocessing pipeline evaluation [6,12,14,39]

Objective 2: Transfer Learning Comparison

To compare the classification performance of multiple pre-trained CNN architectures—including ResNet-50, DenseNet-201, VGG-16, MobileNetV2, and InceptionV3—fine-tuned via transfer learning on the HAM10000 dataset for seven-class skin lesion classification.

Gap Addressed: Limited systematic comparison on HAM10000 [5,7,12]

Objective 3: Data Augmentation for Class Imbalance

To investigate the effectiveness of data augmentation strategies—including geometric transformations, color-based augmentation, and class-specific oversampling—in addressing the severe class imbalance (∼67% nevi) present in the HAM10000 dataset.

Gap Addressed: Class imbalance inadequately addressed [3,7,9,10,11,13,15,33]

Objective 4: Grad-CAM Explainability

To implement Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the spatial regions of dermoscopic images that influence model predictions, thereby enhancing interpretability.

Gap Addressed: Most studies lack interpretability; only Paper [3] uses Grad-CAM on HAM10000 at 82% accuracy

Objective 5: Multi-Metric Evaluation

To assess model performance systematically using a comprehensive suite of evaluation metrics (Accuracy, Precision, Recall, F1-Score, Specificity, AUC-ROC, and Confusion Matrix) rather than relying solely on overall accuracy.

Gap Addressed: Performance metrics often limited to accuracy [7]

Objective 6: Ensemble Methods

To develop a simple ensemble model combining the predictions of top-performing individual architectures to ascertain if ensemble techniques yield superior performance on HAM10000.

Gap Addressed: Robust multi-class classification needed [4,8,15,16,21,22]

Objective 7: Misclassification Analysis

To analyze the specific misclassifications and feature confusion between visually similar classes (specifically melanoma vs. benign keratosis) using confusion matrices and error analysis.

Gap Addressed: Confusion between melanoma and benign keratosis [3,5]

Objective 8: Hyperparameter Tuning

To determine the optimal hyperparameter configurations (learning rate, batch size, optimizer) for fine-tuning deep neural networks on highly skewed dermatological image datasets.

Gap Addressed: Limited systematic comparison of training methodologies [7,12,13]

6. Step 6 — Objective Classification

6.1 Primary Objectives

These represent the core focus and fundamental goals:

Objective	Statement	Rationale
Objective 2	Compare pre-trained CNN architectures via transfer learning	Core experimental activity; establishes baselines
Objective 3	Investigate augmentation strategies for class imbalance	HAM10000's 67% nevi skew demands attention
Objective 5	Comprehensive multi-metric evaluation	Ensures validity beyond misleading accuracy-only

6.2 Secondary Objectives

These enhance and extend the primary findings:

Objective	Statement	Rationale
Objective 1	Evaluate preprocessing pipeline impact	Preparatory step; valuable but not core
Objective 4	Grad-CAM interpretability	High-value addition that sets project apart
Objective 6	Ensemble methods exploration	Optimization after individual model evaluation
Objective 7	Misclassification error analysis	Deep-dive discussion of results
Objective 8	Hyperparameter tuning	Operational step, not novel research objective

7. Step 7 — Objective Scoring and Ranking

Each objective is scored from 1 (lowest) to 5 (highest) across 7 criteria, yielding a maximum of 35 points.

Scoring Criteria

1. Relevance to research topic
2. Connection to literature gap
3. Measurability
4. Experimental feasibility (BCA level)
5. Suitability for HAM10000
6. Academic value
7. Alignment with research questions

Table 10: Objective Scoring Matrix

Objective	Relevance	Lit. Gap	Measurable	Feasibility	HAM10K	Academic	Alignment
forestgreen!10 Obj 2 — Transfer Learning	5	5	5	5	5	5	5
forestgreen!10 Obj 5 — Multi-metric Eval	5	5	5	5	5	5	5
forestgreen!6 Obj 3 — Augmentation	5	5	5	4	5	5	5
forestgreen!6 Obj 4 — Grad-CAM	5	5	5	4	5	5	4
Obj 1 — Preprocessing	4	4	5	4	5	4	4
Obj 7 — Misclassification	4	4	4	5	5	4	4
Obj 6 — Ensemble	4	3	5	3	4	4	3
Obj 8 — Hyperparameters	3	2	5	5	4	2	3

8. Step 8 — Final Selected Objectives

Based on quantitative scoring and systematic mapping to literature gaps, the following objectives are selected:

General / Primary Objective

To develop and evaluate a deep learning-based multiclass skin lesion classification system using the HAM10000 dermoscopic dataset with optimized image preprocessing, data augmentation, and transfer learning techniques to improve classification performance across seven diagnostic categories.

Specific Objective 1 — Preprocessing

To implement and evaluate image preprocessing techniques—including image resizing, normalization, and artifact suppression—on HAM10000 dermoscopic images and assess their impact on downstream classification accuracy.

Addresses Gap: Preprocessing pipeline evaluation [6,12,14,39]

Specific Objective 2 — Data Augmentation

To investigate the effectiveness of data augmentation strategies—including geometric transformations, color-based augmentation, and class-specific oversampling—in addressing the severe class imbalance present in the HAM10000 dataset.

Addresses Gap: Class imbalance [3,7,9,10,11,13,15,33]

Specific Objective 3 — Transfer Learning Comparison

To compare the classification performance of multiple pre-trained CNN architectures—including ResNet-50, DenseNet-201, VGG-16, MobileNetV2, and InceptionV3—fine-tuned via transfer learning on the HAM10000 dataset.

Addresses Gap: Limited systematic comparison on HAM10000 [5,7,12]

Specific Objective 4 — Multi-Metric Evaluation

To analyze the per-class classification performance across all seven HAM10000 diagnostic categories using comprehensive evaluation metrics including accuracy, precision, recall, F1-score, sensitivity, specificity, AUC-ROC, and confusion matrix analysis.

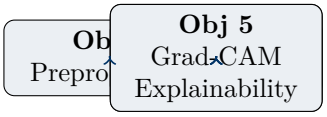
Addresses Gap: Metrics limited to accuracy [7]; confusion between similar classes [3,5]

Specific Objective 5 — Grad-CAM Explainability

To assess the interpretability of the best-performing deep learning model using Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize and validate the discriminative lesion regions used for classification decisions.

Addresses Gap: Only Paper [3] uses Grad-CAM on HAM10000 at 82% accuracy

8.1 Logical Pipeline



9. Step 9 — Objectives to Research Questions Mapping

Table 11: Research Objective to Research Question Mapping

Research Objective	**Research Question**	**Experiment**	**Dataset**	**Eval. Metrics**	**Expected Outcome**
Obj 1: Implement and evaluate preprocessing techniques	**RQ1:** How do preprocessing techniques affect classification accuracy on HAM10000?	Compare baseline models on raw vs. preprocessed images (DullRazor + CLAHE)	HAM10000	Accuracy; Loss	Preprocessed images yield 2–5% improvement in accuracy and faster convergence
Obj 2: Investigate augmentation for class imbalance	**RQ2:** What augmentation strategies best mitigate class imbalance?	Train identical CNNs on non-augmented vs. class-balanced augmented data	HAM10000	F1-Score; Class-wise Recall	Balanced augmentation improves minority class recognition
Obj 3: Compare pre-trained CNN architectures	**RQ3:** Which pre-trained CNN achieves highest performance via transfer learning?	Fine-tune ResNet-50, DenseNet-201, VGG-16, MobileNetV2, InceptionV3 under identical conditions	HAM10000	Accuracy; AUC-ROC; F1-Score	DenseNet-201 or ResNet-50 exceeds 90% accuracy
Obj 4: Analyze per-class performance	**RQ4:** How does performance vary across the 7 categories?	Extract classification reports and confusion matrices for best model	HAM10000	Per-class Precision, Recall, F1, Specificity	High *nv* accuracy; *mel*–*bkl* misclassification identified
Obj 5: Assess interpretability via Grad-CAM	**RQ5:** Can Grad-CAM produce clinically meaningful activation maps?	Generate Grad-CAM heatmaps for correct and incorrect predictions	HAM10000	Qualitative visual assessment	Model focuses on lesion morphology, not artifacts

10. Step 10 — Objectives to Project Execution Mapping

Table 12: Objective to Practical Execution Flow

Objective	Method	Dataset	Model	Experiment	Eval. Metric	Expected Result
Obj 1: Preprocessing	DullRazor hair removal; CLAHE contrast; normalize [0,1]	HAM10000 (10,015)	Baseline MobileNetV2	Train on raw vs. preprocessed; compare curves	Accuracy; Val Loss	2–5% accuracy boost; faster convergence
Obj 2: Augmentation	Geometric transforms; selective minority oversampling	HAM10000	Baseline CNN	With vs. without balanced augmentation	Macro F1; Class Recall	Macro F1 increases; minority detection improves
Obj 3: TL Comparison	Transfer learning; ImageNet weights; custom dense heads	HAM10000 (prep. + aug.)	ResNet-50, DenseNet-201, VGG-16, MobileNetV2, InceptionV3	Train all 5 under identical conditions; rank	Accuracy; AUC-ROC	Best model >90% accuracy
Obj 4: Per-class Analysis	Classification reports; confusion matrix decomposition	HAM10000 (test split)	Best from Obj 3	Predict on test set; per-class metrics	Precision, Recall, F1, Specificity	Identify mel–bkl confusion cluster
Obj 5: Grad-CAM	Extract gradients from final conv layer; overlay heatmaps	HAM10000 (sample)	Best from Obj 3	Correct & incorrect prediction heatmaps	Visual assessment	Model focuses on lesion, not background

11. Step 11 — Final IEEE-Style Research Objectives

Research Objectives

General Objective

General Objective

To develop and evaluate a deep learning-based multiclass skin lesion classification system using the HAM10000 dermoscopic dataset with optimized image preprocessing, data augmentation, and transfer learning techniques to improve classification performance across seven diagnostic categories.

Specific Objectives

1. **To implement and evaluate** image preprocessing techniques—including image resizing, normalization, and artifact suppression—on HAM10000 dermoscopic images and assess their impact on downstream classification accuracy.

2. **To investigate** the effectiveness of data augmentation strategies—including geometric transformations, color-based augmentation, and class-specific oversampling—in addressing the severe class imbalance present in the HAM10000 dataset.

3. **To compare** the classification performance of multiple pre-trained CNN architectures—including ResNet-50, DenseNet-201, VGG-16, MobileNetV2, and InceptionV3—fine-tuned via transfer learning on the HAM10000 dataset for seven-class skin lesion classification.

4. **To analyze** the per-class classification performance across all seven HAM10000 diagnostic categories using comprehensive evaluation metrics including accuracy, precision, recall, F1-score, sensitivity, specificity, AUC-ROC, and confusion matrix analysis.

5. **To assess** the interpretability of the best-performing deep learning model using Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize and validate the discriminative lesion regions used for classification decisions.

Justification from Literature

These objectives directly address critical research gaps identified through systematic analysis of 39 peer-reviewed studies. While several studies have demonstrated that robust preprocessing significantly enhances classification performance [6], [12], [14], [39], the severe class imbalance inherent to the HAM10000 dataset remains the most frequently cited limitation preventing reliable multiclass diagnosis [3], [7], [9]–[11], [13], [15], [33]. By systematically evaluating class-specific data augmentation alongside a comprehensive comparison of modern pre-trained architectures [2], [5], [7], [12], [16], [18], [21], this research seeks to establish an optimized, reproducible classification pipeline.

Furthermore, this study addresses the notable lack of standardized multi-metric evaluation across all seven diagnostic categories [7] and extends upon preliminary explainability efforts [3], [27], [30] by utilizing Grad-CAM to validate that high-performing models make classification decisions based

on clinically relevant morphological features rather than spurious artifacts. The research explicitly positions itself as an academic prototype and does not claim clinical diagnostic capability.

Research Questions

RQ1: How do image preprocessing techniques (resizing, normalization, artifact removal) affect the classification accuracy of deep learning models on HAM10000?

RQ2: What data augmentation strategies are most effective in mitigating class imbalance in the HAM10000 dataset for multiclass skin lesion classification?

RQ3: Which pre-trained CNN architecture achieves the highest classification performance when fine-tuned via transfer learning on HAM10000 for seven-class skin lesion classification?

RQ4: How does classification performance vary across the seven diagnostic categories of HAM10000, and which categories are most frequently misclassified?

RQ5: Can Grad-CAM visualizations from the best-performing model produce clinically meaningful activation maps that highlight diagnostically relevant lesion regions?

Research Gap to Objective Mapping

Table 13: Mapping of Research Gaps to Objectives

Research Gap	Supporting Papers	Addressed By
Class imbalance inadequately addressed	[3],[7],[9],[10],[11],[13],[33]	Specific Objective 2
Limited TL comparison on HAM10000	[5],[7],[12]	Specific Objective 3
Insufficient preprocessing evaluation	[6],[12],[14],[39]	Specific Objective 1
Most studies lack explainability	[3] (only Grad-CAM)	Specific Objective 5
Metrics limited to accuracy only	[7]	Specific Objective 4
Confusion between similar classes	[3],[5]	Specific Objective 4

References

Paper numbers [1]–[39] correspond to the 39 research papers analyzed in this document, indexed according to `papers_metadata_index_39.csv` and cited via `all_39_journals.bib`. The full bibliographic entries are maintained in the companion BibTeX file.

This document was generated as part of the Research Methodology course (BCA 7th Semester). All findings are evidence-bound to the 39 analyzed papers. No claims are fabricated.